

Muhammad Hasan Arsalan

+92-305-7059468 | ✉ hasanarsalan781@gmail.com | in hasan-arslan-71bb14220 | GitHub HasanCT282 | 📍 Karachi, Pakistan

EDUCATION

NED University of Engineering & Technology

Bachelor of Science in Computer Science (BSCS) — CGPA: 3.71 / 4.00

2024 – Present

Karachi, Pakistan

Govt. Delhi College

Higher Secondary School Certificate (HSSC) – Pre-Engineering — Marks: 85%

2022 – 2024

Karachi, Pakistan

Montessori Complex High School

Secondary School Certificate (SSC) — Marks: 92%

2020 – 2022

Karachi, Pakistan

PROJECTS

Flight Delay Analysis & Prediction System | Python (Streamlit), MySQL

- Built a **real-world database system** that ingests and analyzes public flight datasets to track and predict delays, organizing data across normalized tables for flights, airlines, airports, and daily weather.
- Implemented **database triggers** to automatically flag airlines exceeding a delay threshold at a specific airport within a single day, enabling proactive anomaly detection without manual querying.
- Leveraged **stored procedures, views, and indexing** on a massive real-world dataset to optimize query performance and surface actionable delay patterns through an interactive Streamlit dashboard.

Student Management System | Qt Framework, C++

- Engineered a fully functional **desktop GUI application** using Qt Framework and C++, enabling seamless management of student enrollment, records, and academic data for institutional use.
- Implemented comprehensive **CRUD functionality** — add, edit, delete, and search — within a clean, user-friendly interface that significantly reduces administrative workload.
- Applied **Object-Oriented Programming (OOP)** principles to produce a modular, scalable, and maintainable codebase ready for future feature expansion.

Electricity Management System | DSA, Console Application

- Designed and deployed a **console-based billing and usage-tracking system** leveraging core Data Structures & Algorithms to ensure efficient consumer data management and billing accuracy.
- Structured consumer records and usage data using appropriate **DSA constructs**, demonstrating practical translation of theoretical concepts into reliable, real-world solutions.
- Delivered clean, well-documented code architecture showcasing strong **analytical and logical thinking** capabilities applicable to complex software engineering challenges.

SKILLS

Languages : C, C++, Python

Tools & Technologies : Visual Studio Code, Dev C++, Git & GitHub, Qt Framework, MySQL, Streamlit, MS Office Suite, Graphic Design Tools

Core CS Concepts : Data Structures & Algorithms, Object-Oriented Programming, Database Design & Normalization, Problem Solving, Logical Thinking

ACHIEVEMENTS

- Maintained an outstanding **CGPA of 3.71/4.00** at one of Pakistan's most rigorous engineering programs (NEDUET BSCS).
- Achieved **92% in SSC** and **85% in HSSC** (Pre-Engineering) — consistently placing among top performers across all academic levels.
- Successfully delivered **three complete software projects** (database system, GUI app, and console app) within academic timelines, demonstrating project ownership and execution capability.
- Developed foundational **graphic design skills** to complement technical expertise, enabling effective visual communication of ideas.

LANGUAGES

Urdu : Native / Mother Tongue

English : Professional Working Proficiency